

- Built execution patterns that routed Prefect workloads to EC2 or Fargate based on runtime conditions such as file size and processing requirements.
- Created EC2- and Fargate-backed microservices for OCR, document conversion, data extraction, and backend processing workflows.
- Designed and maintained REST APIs and message queue architecture for high-load document-processing services, improving resilience and throughput in production data exchange.
- Automated build, test, and deployment workflows for Python and JavaScript services using GitHub Actions and AWS CodePipeline.

Data Engineer

2019–2022

Oak Ridge National Laboratory

Oak Ridge, TN

- Replaced a brittle cron-based web scraping process with a more reliable Airflow ETL workflow deployed in Kubernetes and cloud infrastructure.
- Developed full-stack Docker images and automated spatial-temporal data acquisition workflows using Python, R, Airflow, and Shiny for WSTAMP and Hadoop-backed data storage.
- Led development for a COVID-19 tracking project spanning R, Python, geospatial analysis, downstream model creation, and operational reporting workflows; received distinguished recognition for the project.
- Built containerized Airflow ETL workflows, Python microservices, and Swagger/OpenAPI REST APIs for NAERM-RTSA and related data exchange with PostgreSQL, AWS S3, AWS RDS, and MongoDB.
- Automated web scraping and social media ingestion for misinformation classification, statistical modeling, network analysis, Elasticsearch indexing, and Kibana visualization.
- Served as Software Quality Assurance board chair for the Geospatial Science and Human Security Division and contributed to ORNL Data Council discussions on data management, governance, and architecture.

Data Scientist II Subcontractor

2018–2019

Oak Ridge National Laboratory

Oak Ridge, TN

- Automated WSTAMP web application data ingestion using R, Python, Airflow, and Docker.
- Authored an R package to normalize geographic identifiers and unify data access across online APIs.
- Developed a Spatialite-backed SQLite database to simplify lookup, retrieval, and integration of geographic data.
- Built a Python zonal statistics application for calculating raster summaries across shapefile geometries.

Owner & Contractor-For-Hire

2018

Specrabella

Knoxville, TN

- Designed and deployed interactive energy informatics applications using Leaflet, Shiny, Snowflake, and machine-learning-derived calculations.
- Consulted on greenfield web application architecture, database design, visualization, and deployment for energy-sector clients.
- Built data ingestion and visualization workflows using cURL, PHP, R, Python, and Snowflake for energy datasets.
- Designed PostgreSQL databases and development workflows for energy and financial data products.

Bioinformatician, Project Manager, & Technical Sales Representative

2016–2018

Microbial Insights

Knoxville, TN

- Established an automated NGS analysis pipeline using R, MySQL, Python, Bash, JavaScript, and Shiny-backed internal tooling.
- Automated client-facing statistical workflows including linear models, ANOVA, self-organizing maps, clustering, PCA, and PCoA.
- Developed ETL and customer data visualization tooling using R, PHP, jQuery, and MySQL for dynamic qPCR result analysis.
- Built a custom R package, local Shiny applications, and an internal PostgreSQL/PHP wiki to improve reporting, project management, and customer service workflows.

Graduate Research Assistant

2014–2016

Dr. Neal Stewart's Plant Biotechnology Lab, University of Tennessee

Knoxville, TN

- Analyzed cell suspension experiments using linear models, ANOVA, PCA, Python, and R for DOE reporting and scientific communication.
- Evaluated suspension cultures through chemical and spectral processes and translated findings into research deliverables.
- Collaborated on a novel single-cell suspension and cryopreservation robotic system while executing monocot genetic modification workflows.

Laboratory Assistant

2012–2014

Dr. Neal Stewart's Plant Biotechnology Lab, University of Tennessee

Knoxville, TN

- Developed Python-based statistical methodology for screening lignin content and imaging cell characteristics both *in vivo* and *in vitro*.

